The Cooperative Scope of Local Gods: Punishment, Omniscience, and Distant Cooperation among the Sihanaka of Madagascar

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Abstract

Do supernatural beliefs in punishment and omniscience promote cooperation among distant unrelated individuals? Previous cross-cultural research – predominantly among "moralistic" religious traditions – suggests it might, although numerous open questions remain. Foremost among these are: "Do these results replicate among *local* religious traditions?" and "Is the relationship causal?" Here, we take a causally-informed approach to explore these questions among the Sihanaka from Madagascar. Our research question is: Do punishment and omniscience beliefs about a local god - *zazavavindrano*, an aquatic deity - promote cooperation towards co-religionists of this tradition? We assessed cooperation using Random Allocation and Dictator Games. Bayesian regression models were used, adjusting for plausible confounding factors, with *g*-computation methods to estimate our counterfactual causal effects of interest (i.e., the joint effect of *zazavavindrano* punishment and omniscience beliefs on cooperation with distant co-religionists). Across all games, conditions, and confounding assumptions, we found little evidence for a relationship between these *zazavavindrano* beliefs and cooperation. While *zazavavindrano* beliefs may not promote cooperation with unknown distant co-religionists, whether these beliefs facilitate cooperation in other domains – such as ecological management of wetlands, which is of greater concern to *zazavavindrano* than moral behaviour towards strangers – is an interesting open question for future research.

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1. Background

A range of convergent theories associate religious beliefs and practices with cooperation and the kinds of solidarity that bolster a relative stability present in many societies. While these arguments date back to the foundational work of Durkheim (2001 [1915]) and Malinowski (1992 [1948]), their core insights have been refined by later work such as Rappaport (1999) and remains influential today. Some contemporary work suggests that, despite considerable cross-cultural variation, beliefs in supernatural punishment might have contributed to minimizing defection in cooperative relationships (Johnson, 2005; Schloss & Murray, 2011). Such beliefs may have even increased the scope of those relationships by fostering a sense of more general applicability (Norenzayan et al., 2016). Specifically, beliefs in gods that are maximally morally concerned, punitive, and knowledgeable may be especially effective in galvanizing the kinds of decisions necessary to foment cooperative bonds across individuals in anonymous interactions or in contexts without kinship or repeated interactions to structure cooperation. It is precisely this kind of cooperation that is assumed to be necessary for the kinds of social complexity that are exclusive to humans.

According to this work, what partially accounts for the current global coverage of the so-called "moralistic traditions" (e.g., the Abrahamic and Dharmic traditions) was this ability to harness cooperation and propel such behavior beyond parochial confines. The expanded cooperation they enjoyed facilitated alliances and wider trade networks that put such groups in a better position to incorporate, compete with, and dominate other groups. Yet, social scientists have debated the global ubiquity and utility of the relationship between religion and morality for generations (Purzycki, 2025b), leaving an open question: *are individuals in the "moralistic" traditions – those with beliefs about gods' moral punishment – more likely to cooperate beyond local boundaries than traditions that revolve around gods that care about things other than morality?* Put differently: *are the cooperative effects of "local" traditions – those with gods that are more concerned with other human activities – more confined to local context than "moralistic traditions?"*

While group-level studies show notable variation in the relationship between cooperation and moralistic supernatural punishment (Beheim et al., 2021; Turchin et al., 2023; Watts et al., 2015), considerable cross-cultural work shows that, on average, individuals who claim their gods punish and know about human activities are more likely to be generous and impartially fair toward anonymous, geographically distant co-religionists who are incapable of reciprocating than those who claim otherwise (Lang et al., 2023; Purzycki et al., 2024). However, at best, this previous empirical work showed a stable effect of beliefs about gods' punishment and knowledge breadth, but their attributed moral concerns did *not* show a similar effect (Bendixen et al., 2023; Purzycki, Lang, et al., 2022). In other words, while gods' punishment and knowledge might boost cooperation beyond parochial boundaries, their specific moral concerns

do not appear to do much of the work.[2] Here, we examine the relationship between the breadth of cooperation and the traditional belief sets of a Sihanaka community in Madagascar.

2. Ethnographic Background of Field Site

2.1 The Sihanaka

The Sihanaka, meaning "people of the marsh," number over half a million and are among Madagascar's oldest of the eighteen officially recognized ethnic groups (Norge, 2025). They inhabit the fertile Alaotra basin in the island's central northeast, a lacustrine landscape of marshes, extensive rice plains, and Madagascar's largest freshwater body – Lake Alaotra. This "granary of Madagascar" supports livelihoods based on irrigated riziculture, fishing, and the keeping of zebu cattle and domestic fowl such as geese and ducks. Over centuries, Sihanaka culture and cosmology have developed in close symbiosis with the wetland ecology, with riziculture and ritual structured by the annual cycle of the rainy season (November–April) and the dry season (May–October) as well as calendrical systems of Sanskrit origin.

Although the Alaotra basin has received little direct archaeological investigation, its agricultural potential and position at the head of the Maningory River – which forms a natural corridor to the Indian Ocean coast – make it a compelling candidate for early settlement. Given that habitation extended into the central highlands by the eleventh century, it is reasonable to propose that the basin was first inhabited in the early second millennium CE by Afro-Austronesian migrants moving inland from the east coast. These early settlers adapted Austronesian fertility traditions while venerating metahuman beings of the wetlands such as the *zazavavindrano* (discussed below). From the late thirteenth century, immigrants likely of Javanese descent arrived via the northeast coast, bringing irrigated riziculture and the nobility title *andriana* (from Javanese *rahadyan*), which reinforced social differentiation between free landowners (*andriana*) and enslaved labourers (*andevo*) (Norge, 2025).

At least by the seventeenth century, Sihanaka polities had become powerful actors in Madagascar's interior. The conversion of the Alaotra marshes into irrigated paddy fields, aided by irrigation techniques attributed to Javanese influence, generated agricultural surpluses that sustained population growth and military expansion. Federations of villages under semi-divine leaders known as *mpanazary* organized large-scale slave raids against neighboring populations on the eastern coast and in the central highlands. Captives, together with herds of zebu and baskets of rice, were marched northward to Arabo-Swahili slave ports such as Langany and New Masselage on the northwest coast, where they were sold to Arabo-Swahili and later European ships. These exchanges brought silver, firearms, and imported textiles, fueling both prosperity and warfare. In the eighteenth century, conflict with the rising Sakalava kingdom of Boina

[2] Recent efforts (Bendixen et al., 2024; Bendixen & Purzycki, 2020) suggest that this particular domain of religious traditions – cultural models of gods' concerns – might be more of a reflection of social coordination where the concerns of gods revolve around behaviors that constitute cooperative options in locally salient social dilemmas. As such, they are more appeals to influence others' behavior rather than motivations of individual behavior (cf. Fitouchi et al., 2025).

curtailed Sihanaka expansion but also produced deep cultural fusion: Sihanaka rulers appropriated Sakalava royal rituals and symbols, blending them with older fertility rites that remain visible in shrine practice today. In 1823, the Sihanaka region was conquered by the expanding Merina state, whose rulers had entered cooperation with Great Britain that restricted the slave trade and opened the kingdom to Christian missions and technological modernization. The Sihanaka were disarmed and made subjects of the Merina monarchy based in Antananarivo. Protestantism began spreading from the late 1860s, followed later by Catholicism, taking strongest root in hilltop towns such as Ambatondrazaka and Amparafaravola on the fringes of the Alaotra basin, where mixed Merina–Sihanaka populations resided. Marshland villages such as Anororo (the main fieldsite of the present research), by contrast, remained relatively isolated within the wetlands and continued to preserve Sihanaka institutions of ritual authority and collective taboos (*fady*; Razafimbelo, 1978, 1984).

As France colonized Madagascar in 1896, large tracts of ancestral Sihanaka land were expropriated and granted as agricultural concessions to European settlers. As of the 1940s, these concessions became sites of mechanized rice cultivation, further intensifying production and continuing the long transformation of marshland into an engineered landscape of paddy fields, canals, and dams. After independence in 1960, political power once again centralized in Antananarivo, but the *mpanazary* realms endured through local ritual orders; in several areas once ruled by a *mpanazary*, annual fertility rituals and the everyday observance of ancient taboos (*fady*) sustain the socio-cosmology order of the *mpanazary*.

2.2 Spiritual Landscape

2.2.1. General

Sihanaka cosmology is grounded in an immanentist ontology (Sahlins, 2022; Strathern, 2019) in which all realms of existence participate in a single continuous cosmos animated by the divine power of *hasina* (Norge, 2025). Comparable to Oceanian *mana* or Indonesian *semangat*, *hasina* circulates through plants, water, and words, and can be accumulated, preserved, and transmitted through ritual action. Rather than dividing nature from supernature, the cosmos is conceived as a graded continuum of beings, places and objects endowed with different degrees of *hasina*, the sacred force that animates all life. Those most saturated with *hasina* appear invisible to ordinary perception – not because they dwell in a separate metaphysical otherworld, but because their potency exceeds human sensory capacity. Unlike transcendentalist religions such as Christianity, where divine power is ontologically transcendent and external to the world, Sihanaka thought locates it within the world itself: *hasina* emanates from the celestial creator Zanahary and circulates as immanence throughout the cosmos, sustained and directed through ritual practice.

For the Sihanaka, the Alaotra basin is an enchanted landscape of multiple realms inhabited by beings of varying *hasina*: human villages bound by kinship and agriculture, tombs of the *razana* (ancestors), shrines (*doany*) housing divinised *mpanazary* (precolonial diviner-rulers), and the submerged settlements of *zazavavindrano* (aquatic humanoids) beneath Lake Alaotra. Each realm sustains and channels vitality in its own way, together forming a

cosmopolitical world where humans and divinities interact across permeable boundaries. Through ritual, people engage the *zanahary* to secure rain, ensure fertile rice, or protect the living and their livestock. Some rituals are collective – such as shrine ceremonies for Ndrianampanjaka – while others are intimate healing rites held in the homes of spirit mediums. In both, collective concerns intertwine with individual supplications: community fertility and rain are sought alongside requests for help with infertility, marital discord, or sickness.

The Sihanaka pantheon is open, dynamic, and regionally varied. At its apex stands the celestial creator Zanahary (also Andriamanitra), source of all *hasina*, from whom emanate numerous lesser divinities – collectively also termed *zanahary,* with a lower-case "z" – that populate the world's manifold domains. Most are metahuman in Sahlins's (2022) sense: beings who share the perceptual, affective, and bodily capacities of humans, yet in heightened form through the intensification of *hasina*, enabling powers of healing, prophecy, and invisibility. They care deeply about behaviour: the *razana* oversee kinship morality; *mpanazary* uphold ritual and social order; and *zazavavindrano* guard the purity of water and relational harmony between humans and their aquatic beings.

Foremost among these are the *zazavavindrano* – the "water people" and the "local" god in our study – who dwell in the marshes and depths of Lake Alaotra. Renowned for their luminous beauty and long hair, they inhabit underwater villages with houses, fields, and zebus mirroring human life. Guardians of their watery domains, they reward purity, respect, and quiet ritual observance while punishing pollution, betrayal, and disorder with illness, infertility, or madness. Relations between humans and *zazavavindrano* vary: they appear in dreams or trances, intermarry with humans, or communicate through mediums who channel them in healing ceremonies.

Another major category of *zanahary* comprises the *mpanazary* – precolonial diviner-rulers celebrated in oral tradition for their thaumaturgical powers: walking across Lake Alaotra on water-lily leaves, curing epidemics, or commanding hailstorms. Rather than dying, they became immortal rulers residing invisibly in communal shrines (*doany*). In Anororo, Ndrianampanjaka (the "moralistic" god in our study) and his successors Andriatsimitoviaminandriana and Mahatolotrarivo continue to govern their ancestral realm through ritual order and taboo. Ndrianampanjaka values respect, loyalty, and unity, while his anger is provoked by disrespect, impurity, or transgression. His order is embedded in the rhythm of local life – most notably the twelve-day ritual calendar regulating agricultural work and the enduring collective prohibition against pork.

The *razana* (ancestors) form another influential category of *zanahary*, dwelling in tombs resembling stone houses, sometimes in two storeys with gabled roofs and verandas (Moriyama, 1996). As the enduring members of the kin group (*firazanana*), they maintain moral authority over their descendants through nightly visions (*tsindrimandry*), demanding that ancestral customs be observed and tombs remain well kept, for neglect or violation of ancestral norms brings misfortune (Norge, 2020). Other *zanahary* manifest in animal form, such as the colossal seven-headed serpent Fananimpitoloha said to dwell in Lake Alaotra; ancestors reborn in

crocodilian form; great moths (*Cyligramma disturbans*) regarded as revenants; and the bird Ravolokobokobona, emissary of Zanahary. Ritual prayers (*joro*) also address celestial bodies – the sun, moon, and stars – while older traditions recall divine presences in rice and marsh plants, possibly suggesting an earlier, more animistic kind of immanentism in which personhood extended even more throughout the environment.

The *hasina* of these beings is protected through taboos (*fady*), which define the moral and spatial contours of the Sihanaka world. To be human is to be moral, and to be moral is to respect *fady*. They may apply to a landscape feature (a marsh where salt is forbidden), a lineage (which abstains from onions or chicken), or a sacred object (an amulet intolerant of certain foods). Because *zanahary* are situated beings with heightened perception, they detect violations swiftly. Transgressions against *fady* are punished according to scale: *fitaroka fohy* ("short flogging") denotes individual misfortune – an illness, accident, or death – whereas *fitaroka lava* ("long flogging") entails collective calamity such as fire, flood, or crop destruction. Villagers recall periodic fires interpreted as punishment for the introduction of pigs into Ndrianampanjaka's territory, or a devastating hailstorm in 2022 attributed to youths gathering shrine timber without the elders' blessing. During possession rituals, participants often plead with the possessing *zanahary* to limit retribution to "short flogging," particularly as Anororo's population has grown more heterogeneous and newcomers may unknowingly violate local *fady*. Each *zanahary* enforces its own moral jurisdiction: Ndrianampanjaka punishes social disorder and impiety with paralysis or death, while the *zazavavindrano* exact more intimate retribution – madness, infertility, or drowning – for pollution or betrayal.

Although Christianity was introduced in the nineteenth century and churches are now widespread, the immanentist premises endure. New entities such as *Jesosy* (Jesus) and *Jehovah* (God, Yahweh) have been incorporated into the cosmology as exceptionally powerful yet worldly *zanahary*. The Christian lexicon itself has been vernacularised: Andriamanitra ("the fragrant one") designates God, and *masina* ("imbued with *hasina*") translates as "holy." The earliest denominations in the area – Protestant, Catholic, and Lutheran – were largely absorbed into Sihanaka cosmology, whereas newer evangelical churches have more forcefully opposed ancestral ritual, condemning possession and shrine worship as demonic. The relationship between Christianity and traditional cosmology remains ambivalent: Christianity is associated with modernity and education, while ritual practice is sometimes framed as backward – a hierarchy occasionally internalised by Sihanaka themselves. Yet this coexistence also fosters quiet pride: the endurance of ancestral ritual is seen as something distinctly and authentically Sihanaka, a moral heritage that continues to define belonging in the region.

In the heart of the Sihanaka world, along the shores of Lake Alaotra, communities such as Anororo continue to sustain a thoroughly immanentist cosmology in which *hasina* circulates between rice fields, tombs, shrines, and people through work and ritual. It remains a distinctly Austronesian system characterised by complementary dualities (Fox, 1987) – sky and earth, male and female, life and death, inside and outside, native and foreign – yet adapted to the wetland

ecology of Alaotra: a "papyrus civilisation" of reciprocal relations between land and people, living and dead, humans and divinities (Norge, 2025).

2.2.2. Ethnographic free-lists and survey

To target the appropriate gods for our study, we conducted preliminary ethnographic interviews that, among other things, entailed freely listing gods (*zanahary* in Malagasy) and answering follow-up questions about those gods. Ethnographic free-lists have the advantage of soliciting naturalistic, ecologically valid data of cultural knowledge (Purzycki, 2025a). With this data, we can quantify and calculate cultural salience, a metric that collapses cultural prevalence and cognitive accessibility into a single metric. Specifically, we use Smith's *S* (Smith & Borgatti, 1997) which is defined as follows:

$$S = \frac{\Sigma(\frac{n + 1 - ki}{n})}{N}$$

where k_i is item order number, n is the list length, and N is the sample size. As such, if every participant listed a particular deity *and* listed it first, it would be $S = 1.00$. To propagate uncertainty around deities' cultural salience scores, we modelled each item type independently in ordered beta-distributed models with uninformative priors (Major-Smith & Purzycki, 2025). We use the AnthroTools package (Purzycki & Jamieson-Lane, 2017) for R to perform all analyses.

After streamlining some of the codes, our sample of 39 individuals listed a total of 41 uniquely specified spirits and gods. The most culturally salient deity was Ndrianampanjaka (Ndriana. in Figure 1), the name of the precolonial *mpanazary* (diviner) considered the "sovereign" (*mpanjaka*) of Anororo – followed by his two *mpanazary* successors Andriatsimitoviaminandriana (Andriat.) and Mahatolotrarivo. After these three *mpanazary*, the most culturally salient divinities are the aquatic metahumans *zazavavindrano* (*zazavavin*.), followed by the ancestors (*razana* in Malagasy), the *mpanazary* Salovambolo (a historical figure associated with Ndrianampanjaka), and finally Andriamanitra and Zanahary, the two names of the distant creator deity.

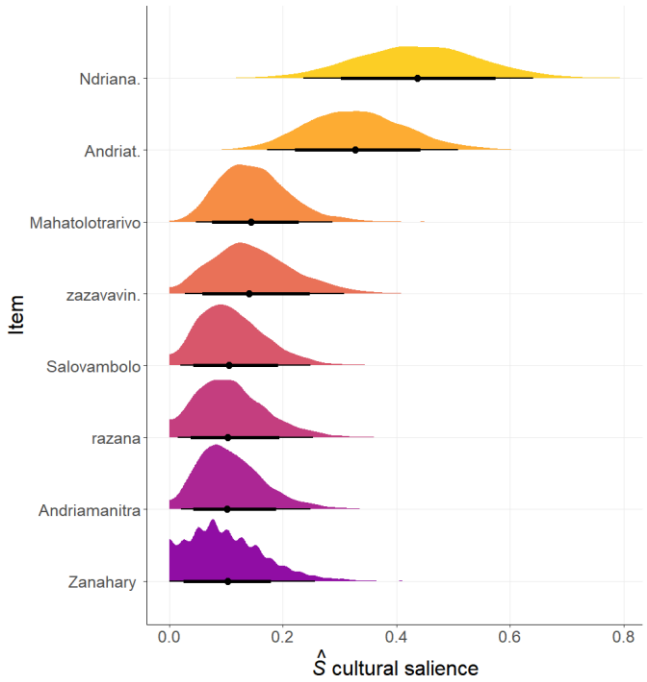


Figure 1. Cultural model of deities. Black circles are Smith's *S* values, with error bars displaying 89% (thick lines) and 95% (thin lines) credible intervals.

Of the 19 who answered follow-up questions about undying sovereign Ndrianampanjaka, all stated that he punished and rewarded people, and virtually all stated that he could see people when they were far away (18) and is concerned with how people behave (17). Of the eight who answered follow-up questions about *zazavavindrano*, all said they reward people, seven said they punish people, and six said they were concerned with how people behaved. Six out of seven claimed they could see what people were doing in far away places. Thus, although Ndrianampanjaka is considerably more salient than the *zazavavindrano*, the data suggest that both are attributed with concern for human behaviour, with punitive and rewarding capacities, and with broad knowledge – although, as noted above, the specific domains and modes of engagement differ.

These patterns of cultural salience and attributed capacities – from precolonial diviners and aquatic metahumans to ancestors and creator gods – align closely with ethnographic findings. Ethnography shows that Sihanaka divinities consistently occupy positions of superior knowledge, and that they participate in the moral and social fabric of Sihanaka life by punishing and rewarding conduct much like humans do – albeit with far greater force and consequence (Norge, 2025). The modelled results therefore mirror, in aggregate form, dynamics that are well attested in long-term ethnographic work.

Based on these results and prior ethnographic research in the Sihanaka region (Norge, 2025), we selected Ndrianampanjaka as the moralistic god and the *zazavavindrano* as the local god for this study. Ndrianampanjaka's role in Sihanaka cosmology aligns closely with the criteria for a moralistic deity: he is understood to be the undying sovereign of the area

surrounding Anororo – corresponding to his precolonial realm – and is strongly associated with social order, adherence to custom, and the protection of the community. These concerns are central to the biannual Feraomby ritual at his royal shrine, a ceremony aimed at securing rainfall and reinforcing social cohesion. Misfortunes such as floods, hailstorms, or town fires are commonly interpreted as expressions of his displeasure, and his high degree of *hasina* (divine power) – linked to his privileged connection to the creator deity Zanahary – is widely taken to confer omniscience.

In contrast, the *zazavavindrano* were chosen as the local god because, although they also punish, reward, and perceive more than ordinary humans, they do so to a markedly lesser degree than Ndrianampanjaka. As aquatic beings considered the masters of the wetlands that dominate the region, they belong to a stratum of divinity whose veneration likely predates ruler-centred ritual traditions – echoing comparative insights that water beings often constitute some of the earliest sacral presences prior to the emergence of more humanised deities (Strang, 2023). Despite their human appearance, *zazavavindrano* remain intimately tied to the watery realm, associated with eels, serpents, and the generative powers of marshland ecology. They are deeply local in both ecological and cultural terms, befitting a people known as the "people of the marsh." Ndrianampanjaka, by contrast, is of foreign origin, stemming from the once powerful Boina Kingdom of the Sakalava on the northwestern littoral. Although firmly rooted in Anororo through his shrine and ritual tradition, his veneration does not extend across the entire Sihanaka region, where other communities maintain their own *mpanazary* rulers in their own shrines.

3. Methods

3.1. Theoretical Model

Our theoretical model of the data-generating process is summarised in the Directed Acyclic Graph (DAG) below (Figure 2; this model was adapted from our core causal model for the wider project, available here). Our estimand – the target quantity of interest we want to estimate (Lundberg et al., 2021) – is the joint causal effect of supernatural beliefs (i.e., beliefs regarding whether *zazavavindrano* are punitive and omniscient) on cooperation with distant co-religionists of the same local god tradition (i.e., villagers in Ambohidava across Lake Alaotra who also cultivate relations with *zazavavindrano*). As data are cross-sectional we do not know the causal relations between the supernatural belief variables, hence focus on their *joint* effect, rather than any individual effects. This DAG makes several assumptions: i) supernatural beliefs cause religious cooperation, and not *vice versa* (i.e., there is no reverse causality); ii) the variables age, sex, education, material security, number of children and fisher occupation plausibly cause both the exposure (supernatural beliefs) and the outcome (religious cooperation), and hence are core confounders of this relationship; iii) there is a risk of unmeasured confounding by additional variables not considered above (denoted by the node "*U*"); and iv) features of the game, such as order of play and comprehension, cause only the outcome, and not the exposure. Based on these assumptions, adjusting for core confounders and assuming no unmeasured confounding, we can estimate an unbiased causal effect of our estimand of interest.

Of course, whether these assumptions – in addition to others required for valid causal inferences such as positivity, consistency and no interference – are met is an open question. For instance, perhaps one of the confounders is in fact a mediator (e.g., supernatural beliefs cause fertility/number of children), in which case any estimate will be biased. Despite this risk, based on previous empirical research and theory we believe that these assumptions are at least plausible for the sociodemographic variables (Baimel et al., 2022; Vardy et al., 2022). We also include fishing occupation as a confounder as fishers plausibly have different beliefs about *zazavavindrano* (as they are a water-based divinity) and may cooperate more with fellow *zazavavindrano* believers. By having a clear causal estimand and making our assumptions clear and discussing how they might be violated, we also facilitate critique and discussion of our causal model, and how different assumptions may impact conclusions.

For instance, although we believe these proposed sociodemographic confounders are plausible in general based on previous research, they may be less relevant in the Sihanaka context specifically. Based on ethnographic observation by AN – who has worked with the Sihanaka since 2015 – supernatural beliefs do not appear to be clearly structured by sociodemographic characteristics such as age, sex or education. Nonetheless, even if in reality there is no relationship between these factors and supernatural beliefs, their inclusion as *potential* confounders is unlikely to result in bias if they are included in statistical models; we can also compare results between models with vs. without these variables as covariates to explore whether the assumption that they are not confounders holds (i.e., if there is no difference in our estimand between these models, this suggests that these sociodemographic factors are not confounders of the focal relationship).

A further factor to consider is the role of potential unmeasured confounders which could bias our causal estimates (denoted by "*U*" in the DAG in Figure 2). Ethnographic research shows that in Sihanaka society, subtle but socially consequential forms of postslavery and autochthony-based stratification continue to shape social relations (Norge, 2025). One dimension concerns *class descent*, namely locally recognised distinctions between descendants of slaves (*andevo*) and free people (*andriana*), which historically structured access to land, ritual authority, and *hasina* (divine power), and which continue to inform perceptions of personhood and belonging, even if rarely spoken of directly. Another dimension concerns *origin descent* – the distinction between long-established "children of the land" (*zana-tany*) and "newcomers" (*vahiny*) – which remains salient in shaping relations to land, wetlands, and the spirits associated with them. Both forms of descent could plausibly influence supernatural beliefs as well as patterns of cooperation with distant co-religionists, thereby acting as confounders. However, because the legacies of slavery and social stratification are among the most sensitive and difficult topics to work with in Madagascar (Evers, 2002; Regnier, 2021), and because such statuses are not openly discussed in Anororo, we did not attempt to collect systematic data on descent. Beyond these considerations, we believe that other major sources of unmeasured confounding are unlikely.

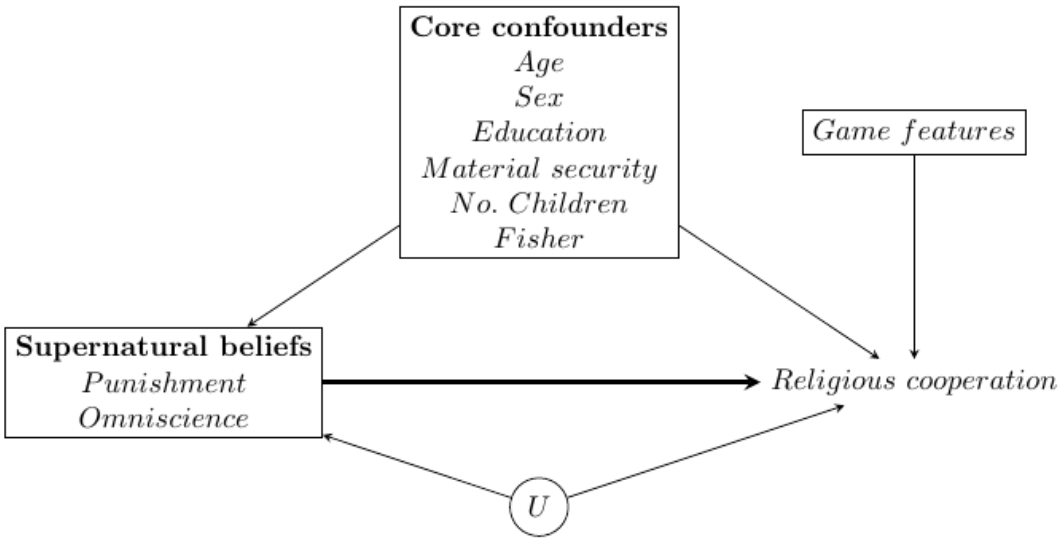


Figure 2: Directed Acyclic Graph (DAG) summarising our causal assumptions regarding the relationship between supernatural beliefs and cooperation with distant co-religionists. The bold arrow denotes our estimand of interest, which is the joint effect of supernatural beliefs on religious cooperation (discussed in more detail in the main text). Causal relations between the core confounders are not displayed here. *U* = Potential unmeasured confounders.

3.2. Materials and Procedure

3.2.1. Procedure

Initial data collection took place in the summer of 2024. A local research assistant conducted interviews with 40 Sihanaka residents of Anororo, administering a short Demographic and Insecurity Survey followed by the Religious Landscape Interview (RLI). These interviews provided ethnographic background on the local religious context and informed the design and interpretation of the subsequent experimental study (as described above).

In the summer of 2025, the field researcher (AN), together with two research assistants, carried out the main experimental and survey components. Participants first completed two behavioural tasks – the Random Allocation Game and the Dictator Game – followed by a 26-item Demographic questionnaire and a 62-item Religiosity questionnaire. These addressed participants' general religiosity, their perceptions of the *mpanazary* Ndrianampanjaka (selected as the moralistic god), the *zazavavindrano* (selected as the local god), and their attitudes toward other social and ethnic groups. A substantial proportion of the 2025 sample consisted of returning participants from the 2024 interview phase.

All materials and instructions were translated into Malagasy by a research assistant, piloted locally, and refined for clarity (available on the project gitHub page here). Each session was conducted in the participants' homes or, occasionally, at the research team's host family's house. Sessions were private, with no other persons present. Because non-literacy is common in this community, participants were given the option to complete the questionnaires orally. In such

cases, a research assistant read each question aloud and recorded the participant's answers verbatim.

Before beginning the games, each participant drew a slip of paper bearing a unique ID number and a randomised game-order code. Gameplay took place on papyrus mats or tables inside participants' houses. Participants put tokens into plastic jugs with lids, and the tokens were small pieces of dried papyrus marked with numerical values. Participants were instructed that these tokens represented real money but were not themselves currency. After completing the session, participants received the corresponding amount in cash. Whenever possible, interviews were conducted immediately after gameplay; when interruptions occurred – for example, funerals or agricultural duties – they were conducted later the same day or, at the latest, on the following day.

3.2.2. Demographics

A number of demographic questions were asked to participants. Of most relevance to this study, this included questions regarding age (in years), sex (male or female), number of years in education, material/food insecurity (an average of four binary questions regarding concerns over running out of food in the next month/6 months/year/5 years), number of children and occupation (coded into "fisher" vs "other" [predominantly farmer]).

3.2.3. Religiosity variables

Participants were asked a battery of questions related to their religiosity, specifically regarding beliefs and practices towards the moralistic god (Ndrianampanjaka) and the local god (*zazavavindrano*). Of most relevance for this study were questions relating to[3]:

1. Punishment: Two questions used in previous research to indicate supernatural punishment were "Do *zazavavindrano* ever punish people for their behaviour?" (yes = 1; no = 0) and "Can *zazavavindrano* influence what happens to people after they die?" (yes = 1; no = 0), with these answers averaged together into a punishment mini-scale (Purzycki et al., 2016). However, among the Sihanaka ideas regarding the afterlife are rather vague and are not something *zazavavindrano* can influence (instead, what is of key importance upon death is entering the family tomb with the ancestors, deities different to those assessed here). As such, approximately half of participants did not answer the second question, and of those that did, only five responded "yes"; for this paper, we will therefore focus on the first "ever punish" variable as our proxy for supernatural punishment.

[3] Note that these questions were designed for consistency as part of a wider cross-cultural project and with previous waves of data collection (Lang et al., 2019; Purzycki et al., 2016). They may therefore lack some ethnographic nuance necessarily relevant to beliefs and practices regarding religion in the Sihanaka region. For instance, what the gods punish for and the importance of where behaviour occurs. While this loss of ethnographic nuance is perhaps not ideal, it is a necessary trade-off with the wider project.

2. Omniscience: Two questions were also used to assess omniscience/monitoring: “Can *zazavavindrano* see into people’s hearts or know their thoughts and feelings?” (yes = 1; no = 0) and “Can *zazavavindrano* see what people are doing if they are far away in Antananarivo (Malagasy capital) [a distant town/city]? (yes = 1; no = 0). These two questions were averaged together into an omniscience mini-scale.

We focus on punishment and omniscience, rather than other supernatural beliefs such as moral concern or reward, for a few main reasons. A first reason is largely pragmatic; previous research has not found moral concern or supernatural rewards to be predictive of such cooperative behaviour (Bendixen et al., 2023; Purzycki et al., 2016). Given that supernatural punishment and omniscience may better motivate cooperative behaviour, we focus on these factors here. A second reason is both conceptual and methodological, as there is ambiguity regarding definitions of “morality” (Purzycki, 2025b) and how to measure it, with previous research using comparable data from other societies finding no relationship between moral concern as measured by free-lists vs scales (Bendixen & Purzycki, 2025).

3.2.4. Behavioral experiments

Two experimental measures of cooperation were used: the Random Allocation Game (RAG) and the Dictator Game (DG). Two versions of both games were played, one with themselves vs. a distant co-religionist (an anonymous recipient from Ambohidava across Lake Alaotra who also cultivates relations with *zazavavindrano*; SELF vs. DISTANT condition), and another with a local co-religionist (anonymous resident of Anororo who also cultivates relations with *zazavavindrano*) vs. a distant co-religionist (LOCAL vs. DISTANT condition). The order of the conditions were counterbalanced, with RAGs always played before the DGs. Participants were given opaque jugs to place allocations in, with each allocation worth 200 ariary (approx. 2% of an average daily wage; meaning the full RAG allocation was worth approximately one day’s wage). Participants were also given a show-up fee of 1,000 ariary (approx. 25% of a day’s wage) for taking part.

For the RAGs, participants were given a fair two-coloured die (three faces black and three faces white) and 30 allocations (tokens made of papyrus with “200” ariary written on them). For each allocation, participants were asked to think which cup they would like to put the allocation in and then roll the die. If the die came up black, they were instructed to put the allocation in the cup they were thinking of, while if it came up white, they should put it in the other cup. If participants played fairly, then on average donations to each cup would be equal; but as participants played by themselves they could easily bias their allocation to whichever cup/recipient they preferred. For the DGs, participants simply divided 10 allocations into each cup as they wished. Comprehension tests were used to ensure that participants understood the rules of the RAGs prior to playing, and that both themselves and recipients would be given the money they allocated.

3.3. Statistical Model

Our statistical model follows our causal model, outlined in the DAG above (Figure 2). That is, our exposure is the joint effect of supernatural punishment and omniscience beliefs in the local gods *zazavavindrano* on the outcome of cooperation towards anonymous distant co-religionists of the same local god tradition. Our core set of covariates to remove confounding bias are age, sex, education, material security, number of children and fisher occupation, with game features (comprehension [RAGs only] and game order) also included as covariates (while game features are not confounders, their inclusion will not introduce bias, and may help improve precision our estimates). As discussed above, it is also plausible that sociodemographic characteristics do not structure supernatural beliefs, and hence are not confounders; we therefore also explored models with only "fisher" as a confounder, to assess whether this impacts our conclusions. Two sets of analyses were conducted for each game, one for the SELF vs. DISTANT condition, and another for the LOCAL vs. DISTANT condition; in both, the outcome is the number of tokens allocated to DISTANT. For the RAGs, we used aggregated binomial models, while ordinal models were used for the DGs.

All analyses were conducted using Bayesian models using Stan in R software (R Core Team, 2024) using the "brms" package (Bürkner, 2017). Four chains of 2,000 iterations (1,000 as warm-up) were used, with visual inspection of trace plots between the chains, r-hat values near 1.00 and effective sample size statistics used to ensure model convergence. Weakly-regularising priors were used.

To estimate the joint effect of supernatural beliefs on cooperation, we used *g*-computation approaches (Hernán & Robins, 2020) to calculate the marginal causal effects of interest (i.e., a change in the punishment and omniscience exposures from both at a value of "0" to both of a value of "1"), based on the posterior predictions from our models. To aid interpretation, these marginal effects are on the count scale (e.g., the change in the number of donations to DISTANT). Exposures were initially treated as independent/additive, but a multiplicative model with an interaction term between the two exposure variables was also tested to see whether it is a better fit to the data.

We used multiple imputation to impute variables with missing data (van Buuren, 2018; White et al., 2011). The largest proportion of missing data for a single variable was 17%, with approximately 30% of cases lost from complete-cases analyses due to missing data (discussed in more detail below). To meet the competing demands of minimising computation time against having enough imputations to sufficiently capture variation in imputed values, we generated 20 imputed datasets. All outcomes, exposures, confounders and covariates detailed above were included in these imputation models, in addition to auxiliary variables to predict missing *zazavavindrano* religious belief values. These auxiliary variables were whether the participant believes in *zazavavindrano*, the frequency they think about and perform rituals towards *zazavavindrano*, as well as all these beliefs regarding Ndrianampanjaka*,* the designated

moralistic god. Based on this imputation model, we anticipate that the imputed data meet the "missing-at-random" assumption required for unbiased imputation. Multiple imputation was performed using the "mice" R package (van Buuren, 2018), with 10 iterations per imputed dataset. These imputed datasets were subsequently analysed using the "brm_multiple" command from the "brms" package to run models across imputations. These imputed results comprise our main analyses, although we also compared these results to those from the complete-case analyses. Note that LOO information criterion statistics for the imputed data have been averaged across each of the imputed datasets.

Example scripts demonstrating the pre-registered analyses on simulated data are available on the project GitHub page here.

3.4. Sampling and Participants

The 79 participants in the 2025 study were recruited through door-to-door recruitment in Anororo. Recruitment aimed to capture a broad cross-section of the local population while maintaining feasibility within the constraints of field-based experimental research. Many participants had previously taken part in the 2024 interview phase, while others were newly recruited.

Participants ranged in age from 17 to 74 and reflected the local occupational composition of the village: most were rice farmers, with others working as fishers, market vendors, weavers, seamstresses, papyrus gatherers, or traditional healers. Approximately equal numbers of both women and men participated, and the sample included individuals of both free (*andriana*) and slave-descendant (*andevo*) backgrounds (although we are unable to link this data to individuals). Participation was voluntary. Some residents declined participation due to other commitments, advanced age, or intoxication.

To reduce the risk of collusion, coordinated responding, or information spillover about the experimental games, recruitment proceeded neighbourhood by neighbourhood. The research team worked intensively within one section of Anororo for several consecutive days before moving to another area on the opposite side of town. When individuals indicated prior knowledge of the games, or when bystanders began to gather during sessions, recruitment in that area was immediately discontinued.

3.5. Pre-registration

A pre-registration document for this paper is available here. Note that as data had already been collected and audited, we published a pre-registration of our *analysis plan*, rather than for *data collection and analysis* (plans for data collection can be found in the grant proposal, available here). When preparing the pre-registration, this knowledge of the data and ethnographic context was useful to help understand some of the complexities of the data and help inform analysis decisions (e.g., patterns of missing data, answering of the afterlife belief questions due to Sihanakan afterlife beliefs, inclusion of "fishing" occupation as a confounder, potential lack of variation in Sihanakan religious beliefs by demographic variables such as age or sex). As also

noted above, we initially developed core causal and statistical models for these site-specific analyses (see here), which we then elaborated upon with the local context in mind for each of the site-specific papers. For full transparency, we note that descriptive statistics for key variables and regressions using this wider project core causal model were conducted in January 2026 for a project workshop, prior to posting the pre-registration, but this was *after* the causal model outlined here had been finalized. Any information from this regarding the data did not therefore influence our causal model or corresponding statistical approach outlined in the pre-registration.

The only analytic difference in this final paper compared to the pre-registration was that the RAG comprehension checks were not included as a covariate in models as there was no variation in responses. While not a deviation from the pre-registration, additional details regarding the multiple imputation procedure have also been given above.

4. Results

4.1. Descriptive Statistics

A summary of the descriptive statistics from both the raw and imputed datasets for the key sociodemographic and religious belief variables (for both *zazavavindrano* and Ndrianampanjaka), along with proportions of missing data, can be found in Table 1. There was little missing data in the sociodemographic variables (max 4 cases; 5% missing), hence the raw and imputed data were largely identical. There was slightly more missing data for Ndrianampanjaka (the moralistic god; 5 to 7 cases; 6% to 9% missing), with more missing for *zazavavindrano* (the local god; 10 to 13 cases; 13% to 17% missing). For both deities, the imputed data suggested somewhat lower punitive and omniscient beliefs compared to the raw data, because participants who did not believe in these deities – and hence did not respond or answered "don't know" to these questions – were less likely to believe in these deities' punitiveness or omniscience. This pattern of non-response towards participants with these beliefs raises the risk of selection bias in complete-case analyses, especially if these beliefs also influence cooperation with distant co-religionists; assuming our imputation models meet the "missing-at-random" assumption, our imputed results ought to be unbiased, however.

Approximately two-thirds of participants believed that *zazavavindrano* punished people for their behaviour (raw = 71.2%; imputed = 65.4%), with more individuals believing that Ndrianampanjaka was punitive (raw = 83.8%; imputed = 80.1%). There was a moderate correlation between these *zazavavindrano* and Ndrianampanjaka punishment beliefs (tetrachoric correlation; raw = 0.57; imputed = 0.51). Similar patterns were observed for the omniscience variables, with the majority of participants answering that *zazavavindrano* could know peoples' thoughts and feelings (raw = 77.6%; imputed = 70.6%) and see what they were doing far away (raw = 71.0%; imputed = 66.3), but these beliefs were slightly more prevalent regarding Ndrianampanjaka (knows thoughts/feelings: raw = 81.9%; imputed = 76.0%: see far away: raw = 75.7%; imputed = 73.4%). There were strong-to-moderate correlations between *zazavavindrano* and Ndrianampanjaka omniscience beliefs (knows thoughts/feelings: raw = 0.57; imputed = 0.68: see far away: raw = 0.75; imputed = 0.51; all tetrachoric correlations), and almost perfect

correlations between responses within each deity (*zazavavindrano* knows thoughts/feeling and sees far away: raw = 0.98; imputed = 0.96: Ndrianampanjaka knows thoughts/feeling and sees far away: raw = 0.91; imputed = 0.89; all tetrachoric correlations). The mean values of the omniscience mini-scale were correspondingly slightly higher for Ndrianampanjaka (raw = 0.80; imputed = 0.75) compared to *zazavavindrano* (raw = 0.77; imputed = 0.68). These *zazavavindrano* and Ndrianampanjaka omniscience mini-scale values were also highly correlated (raw = 0.63; imputed = 0.659; both polychoric correlations).

Table 1: Summary of Sihanaka sociodemographics and religious beliefs regarding punishment and omniscience towards both *zazavavindrano* (local god) and Ndrianampanjaka (moralistic god; *n* = 79). Proportions of missing data for each variable are also presented, along with mean values across the 20 imputed datasets. Note that the cell counts for imputed data can include decimal places, as they are averages.

Variable	N	Raw data – N (%) or median (IQR)	Imputed data – N (%) or median (IQR)	Missing data – N (%)
Age (years)	75	49 (32 to 59)	48.9 (32 to 59)	4 (5.1%)
Sex (male)	77	36 (46.8%)	37.1 (47.0%)	2 (2.5%)
Education (years)	77	8 (5 to 11)	7.85 (4.68 to 11.40)	2 (2.5%)
Number of children	75	3 (1 to 4.5)	3 (1 to 4.72)	4 (5.1%)
Material security	77	0.5 (0.25 to 0.75)	0.5 (0.25 to 0.75)	2 (2.5%)
Fisher occupation	76	10 (13.2%)	11.5 (14.6%)	3 (3.8%)
Zazavavindrano ever punish behaviour	66	47 (71.2%)	51.6 (65.4%)	13 (16.5%)
Zazavavindrano know thoughts and feelings	67	52 (77.6%)	55.8 (70.6%)	12 (15.2%)
Zazavavindrano can see people far away	69	49 (71.0%)	52.4 (66.3%)	10 (12.7%)
Ndrianampanjaka ever punishes behaviour	74	62 (83.8%)	63.2 (80.1%)	5 (6.3%)
Ndrianampanjaka knows thoughts and feelings	72	59 (81.9%)	60.0 (76.0%)	7 (8.9%)
Ndrianampanjaka can see people far away	74	56 (75.7%)	58.0 (73.4%)	5 (6.3%)

As part of the Religiosity Survey, participants were also asked to free-list up to five things each deity likes and dislikes. We coded these items into general codes, such as “Morality,” “Ecology,” and “Ritual” (see Purzycki & McNamara, 2016; summarised in Table A1 in the Appendix).[4] After coding, we calculated Smith’s *S* and 95% uncertainty intervals for all categories across both deities, using ordered beta models (Kubinec, 2022) to generate uncertainty intervals (Major-Smith & Purzycki, 2025). There are clear differences in what these deities care about (Figure 3). For instance, Ndrianampanjaka – the designated moralistic god – is primarily concerned with ritual, with morality being especially salient regarding what this deity dislikes. In contrast, *zazavavindrano* are primarily concerned with food and ecology. While ecological concerns are straightforward – i.e., caring about aquatic life, lake pollution and fishing practices – the code "food" requires some additional context to understand. For the "likes" question, most responses were "fish" and "crab," which are foods eaten by the *zazavavindrano* (although these may also reflect lake-life in general, and hence could be grouped with "ecology"). In contrast, for the "dislikes" question, many participants listed cooked or "human" food, which are *fady* (taboo) as they are considered impure to the *zazavavindrano* and damage their *hasina*, due to the transgression of items between "realms" (i.e., human into *zazavavindrano*). Together, these codes reflect their water-bound environment, with moral concerns being much less salient. As such, our operational definitions of “moralistic” and “local” gods were successful.

In the local RAG condition, participants appeared to allocate tokens fairly (Figure 4), with a mean of 15.1 (SD = 2.9) tokens given to distant co-religionists. An intercept-only aggregated binomial model confirmed this, with no bias in allocations detected (odds ratio = 1.01; 95% credible interval (CI) = 0.93 to 1.10). In contrast, allocations in the self RAG condition were biased more towards self, with a mean of 14.2 (SD = 3.2) tokens given to distant co-religionists; this bias was confirmed in the intercept-only model (odds ratio = 0.90, 95% CI = 0.83 to 0.97). In a repeated-measures aggregated binomial model the bulk of the posterior probability mass indicated that fewer tokens were donated to distant co-religionists in the self compared to the local RAG condition (median difference in predicted allocations = -0.87, 95% CI = -1.89 to 0.18).

For the DGs, tokens were preferentially allocated towards local co-religionists and self (local DG mean = 3.9; SD = 1.7: self DG mean = 3.9; SD = 1.9; Figure 5). From a repeated-measures ordinal model, there was no difference in allocations between the self and local conditions (median difference in predicted allocations = 0.02, 95% CI = -0.42 to 0.46).

[4] We note that these are largely “etic” – rather than “emic” – categorisations, and lose much ethnographic nuance (e.g., which aspects of “Morality,” “Ecology,” or “Ritual” do these deities care about?). However, this classification scheme was used both for consistency with previous publications (Bendixen et al., 2024), and to provide a broad overview on these deities’ concerns; future work can use alternative codings, more appropriate to their research question of interest, if required.

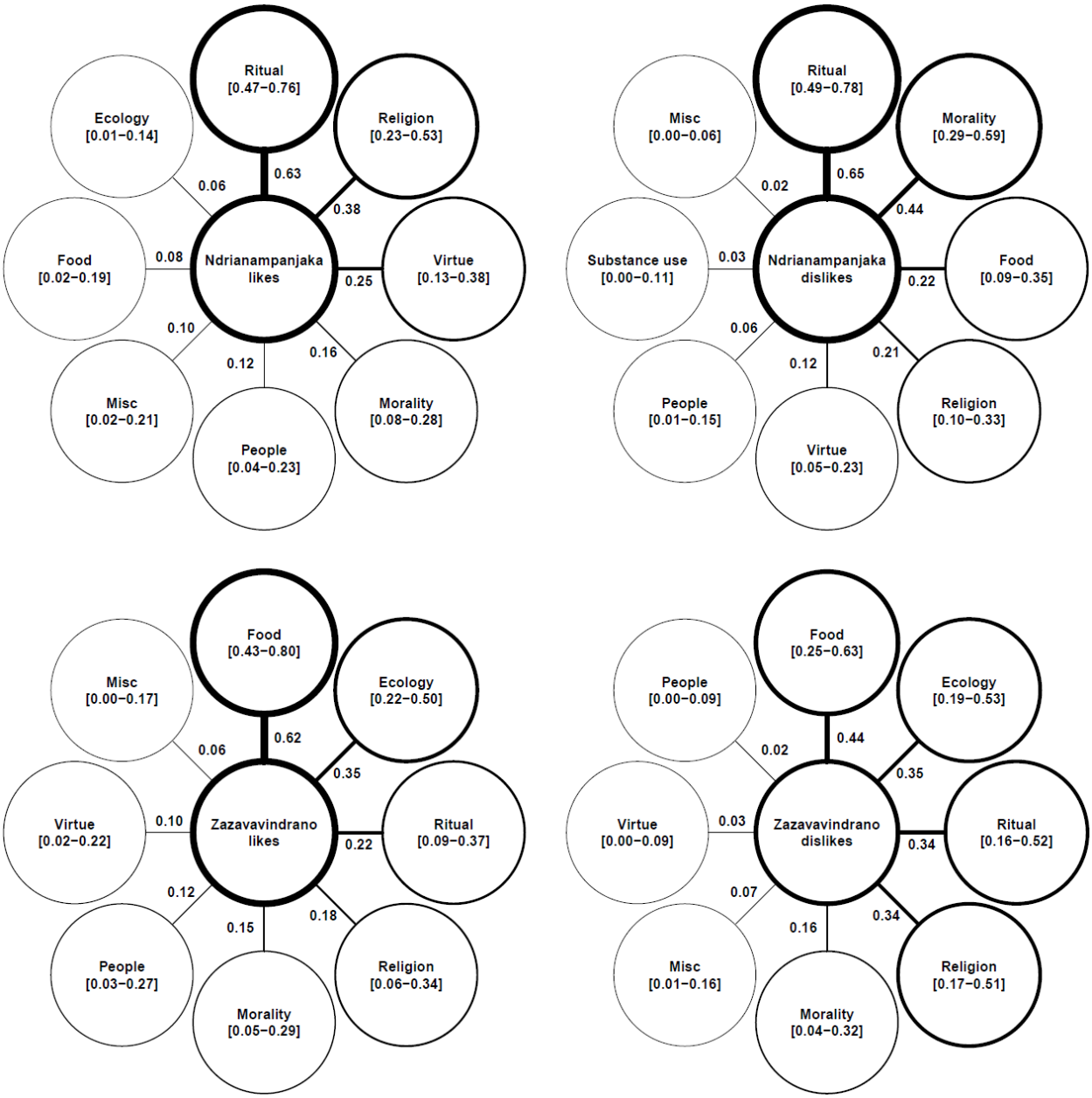


Figure 3: Cultural salience of likes and dislikes for Ndrianampanjaka (the moralistic god) and *zazavavindrano* (the local god). Salience values on the 'stems' are Smith's *S* estimates with 95% credible intervals in the 'petals', based on posterior predictions from ordered beta Bayesian regression models.

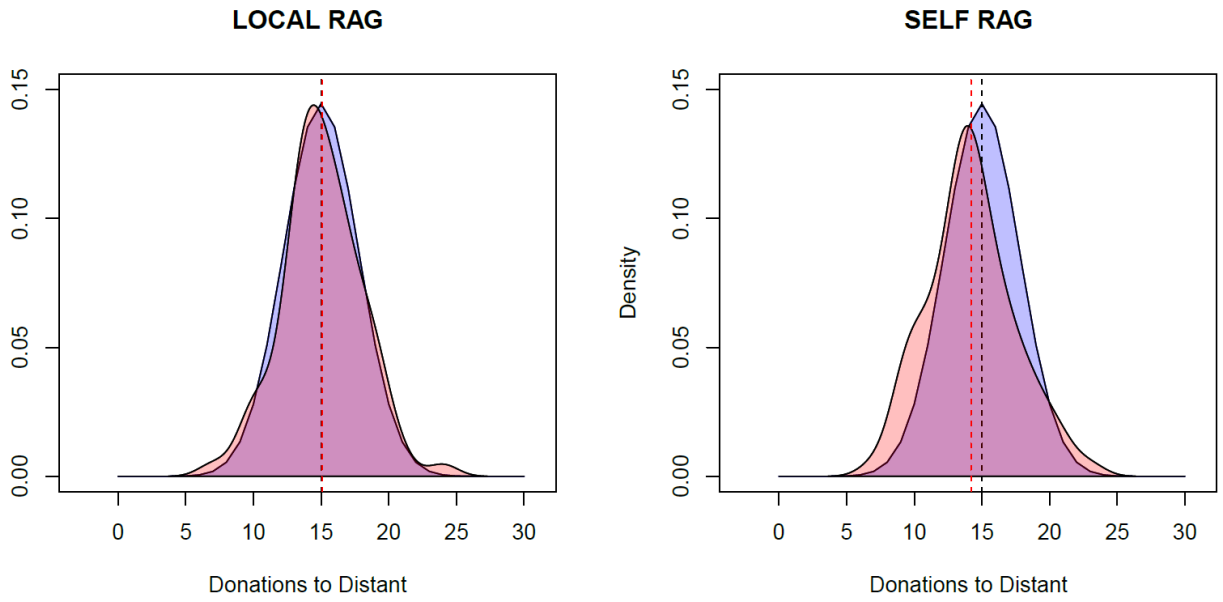


Figure 4: Random Allocation Game (RAG) donations to distant co-religionists in both the local (left) and self (right) conditions. The observed distribution is shaded in red, with the mean observed values indicated by the dashed red line; the theoretical binomial distribution assuming unbiased allocations is shaded in blue, with the mean expected theoretical value indicated by the dashed black line at 15 donations.

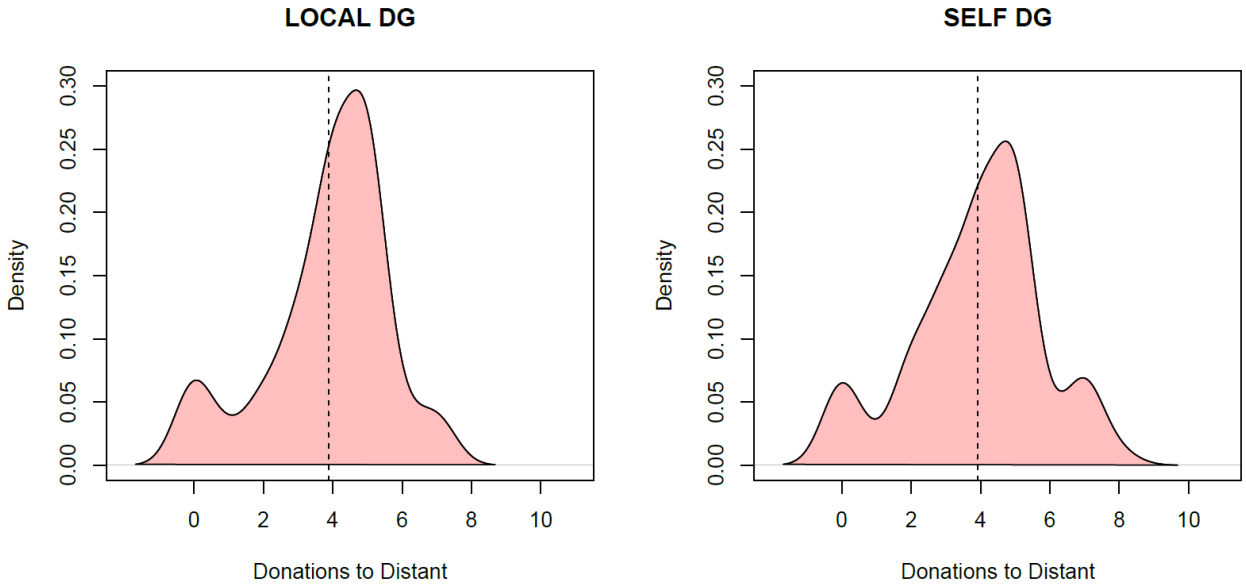


Figure 5: Dictator Game (DG) donations to distant co-religionists in both the local (left) and self (right) conditions. The observed distribution is shaded in red, with the mean observed values indicated by the dashed line.

4.2. Regression Results

The results of our regression and *g*-computation analyses are described next, based on the causal assumptions outlined above and summarised in the DAG in Figure 2. Our main results focus on the multiply-imputed data (n = 79), as these are likely to be less biased and have greater power than the complete-case results. However, as a comparison, we also present the complete-case results, which includes participants with complete exposure, outcome and covariate data across all model specifications (n = 56; 29.1% missing). A summary of all results is given in Table 2.

4.2.1. RAG Results

We first present results using the "core confounders" of Figure 2 (i.e., age, sex, education, number of children, material security and fisher occupation) from the multiply-imputed datasets (Figure 6). For the local RAG condition, there was equivocal evidence for an improvement in model fit when an interaction term between exposures was included; although model fit criteria suggested no improvement (ELPD LOO for additive model = -204.1; ELPD LOO for interaction model = -203.6; ELPD diff = -0.5, SE = 1.7), the 95% credible intervals of the interaction term only just included the null (log-odds estimate = -0.33, 95% CI = -0.79 to 0.13). Given this ambiguity, we therefore present *g*-computation results from both model specifications. In the additive model, there was no relationship between the number of allocations to distant co-religionists between the two counter-factual exposure states (median estimated difference in donations to distant co-religionists = 0.24, 95% CI = -1.71 to 2.20; from approximately 14.9 tokens when punishment and omniscience were both 0, to 15.1 when both were 1). There was little difference in estimates from the interaction model (median diff = 0.48, 95% CI = -1.51 to 2.49; from approximately 14.5 when both 0 to 14.9 when both 1).

In the self RAG condition, there was no evidence for a multiplicative effect of punishment and omniscience beliefs on donations (ELPD LOO for additive model = -213.0; ELPD LOO for interaction model = -215.0; ELPD diff = -1.2, SE = 1.1). Based on this additive model, there was little evidence for a relationship between *zazavavindrano* supernatural beliefs and cooperation with distant co-religionists (median diff = -0.01, 95% CI = -1.90 to 1.87; with an estimated 14.2 allocations at both levels of the exposure).

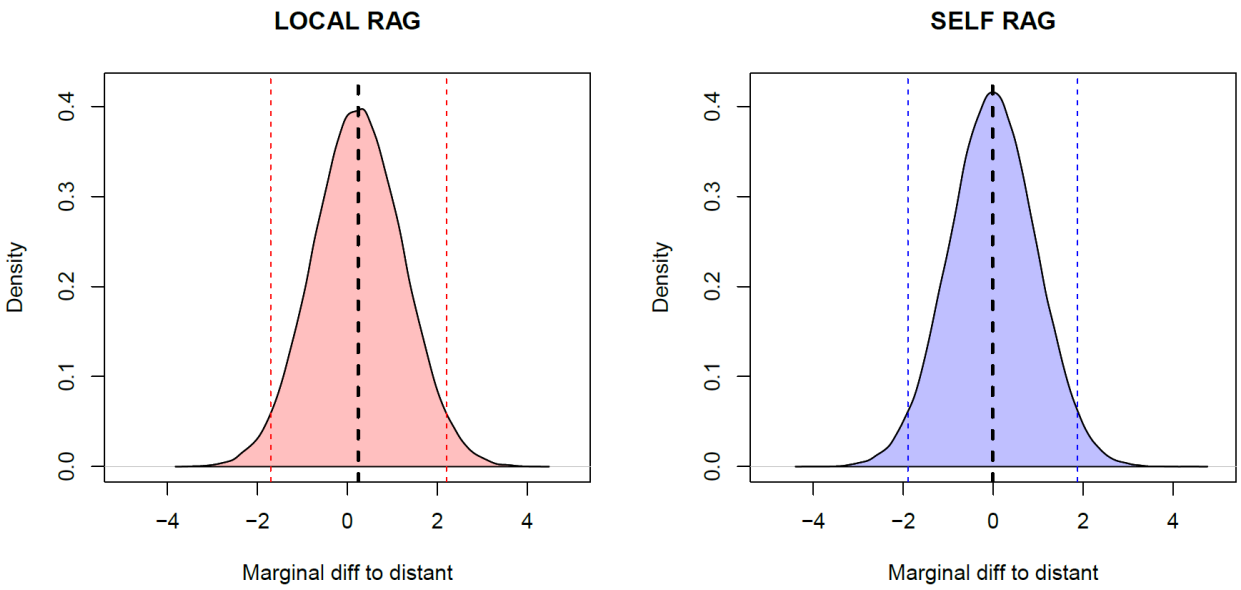


Figure 6: Marginal effect of change in *zazavavindrano* punishment and omniscience beliefs on change in cooperation with distant *zazavavindrano* co-religionist in the Random Allocation Games (RAGs; local RAG = left; self RAG = right). Dashed black lines denote the median expected difference in cooperation (number of tokens) from both punishment and omniscience taking the value "0" to both as "1," with 95% credible intervals in lighter dashed lines.

4.2.2. DG Results

Using the same "core confounders" model for the DG, in the local condition inclusion of an interaction term between exposures did not improve model fit (ELPD LOO for additive model = -142.2, ELPD LOO for interaction model = -141.6; ELPD diff = -0.7, SE = 0.7). The marginal causal contrast from this additive model suggested that increased belief in *zazavavindrano* punishment and omniscience were associated with a slight decrease in donations, albeit with uncertainty intervals which included zero (median diff = -0.42, 95% CI = -1.39 to 0.59; from approximately 4.1 when both 0 to 3.7 when both 1; Figure 7). Results were comparable for the self DG (median diff = -0.39, 95% CI = -1.57 to 0.82; from 4.2 when both 0 to 3.8 when both 1), again based on an additive model, as an interaction term again did not improve model fit (ELPD LOO for additive model = -156.8, ELPD LOO for interaction model = -156.7; ELPD diff = -0.1, SE = 0.4).

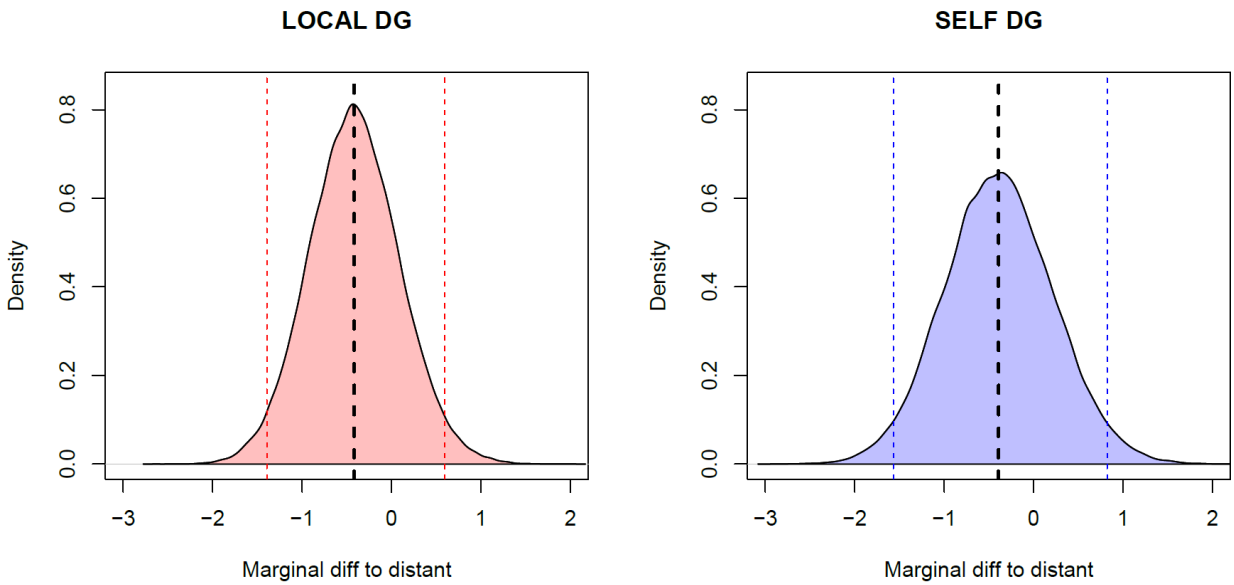


Figure 7: Marginal effect of change in *zazavavindrano* punishment and omniscience beliefs on change in cooperation with distant *zazavavindrano* co-religionist in the Dictator Games (DGs; local DG = left; self DG = right). Dashed black lines denote the median expected difference in cooperation (number of tokens) from both punishment and omniscience taking the value "0" to both as "1," with 95% credible intervals in lighter dashed lines.

4.2.3. Sensitivity Analysis and Complete-Case Results

In the imputed data, the sensitivity analysis which kept just "fisher occupation" as a potential confounder (removing age, sex, education, number of children and material security) gave almost identical results to those reported above (Table 2). That is, the removal of these potential confounders did not impact our estimated causal effect at all, suggesting that these variables were not confounders of our estimand, in line with ethnographic observations (see above).

The complete-case analyses were also broadly consistent to those using the imputed data, with complete-case estimates well within the credible intervals of the imputed estimates (Table 2). The only exception was for the local DG, with the imputed results suggesting a weakly negative relationship and the complete-case results suggesting a potentially positive one, with the complete-case estimates for the local DG outside the previous uncertainty intervals for the imputed data. While it is difficult to separate random variation from systematic selection bias, the difference in results for the local DG does suggest that selection bias may be more likely, although why this bias would be stronger for the local DG data is unclear. As discussed above, the complete-case samples are biased towards participants who believe *zazavavindrano* is both punitive (complete-case sample = 75.9% vs mean of 65.6% in the imputed data) and omniscient (complete-case sample mean of omniscience mini-scale = 0.76 vs mean of 0.68 in the imputed data), meaning that selection bias could be at play here. It is also possible that the "missing-at-random" assumption required for unbiased imputations was not met, meaning that our imputed

results may be biased. While we followed best-practice guidelines for multiple imputation to try and make this assumption more plausible – i.e., including all model covariates, outcomes and additional auxiliary variables to predict missing values of other variables in the imputation model – we cannot rule this out. Equally, both the complete-case and imputation results may be biased. Regardless of the reason(s) for these divergent findings, they do suggest that local DG results are highly contingent and sensitive to the analysis choices and assumptions made. While we would tend to believe that the imputed results are more trustworthy than the complete-case ones, we cannot be certain of this, and should avoid making strong conclusions given these conflicting results. That is, while both of the RAG and self DG results – from both the imputed and complete-case data – are broadly consistent with no strong relationship between *zazavavindrano* beliefs and cooperation with distant co-religionists, the conflicting local DG results are consistent with weakly positive, weakly negative and null relationships.

Table 2: Summary of regression results. All estimates are the counter-factual marginal differences in the number of allocations to distant *zazavavindrano* co-religionists in both the Random Allocation Games (RAGs) and Dictator Games (DGs) comparing between when both punishment and omniscience beliefs in *zazavavindrano* take the values "0" vs "1." These estimates are based on posterior predictions from the corresponding Bayesian model (either aggregate binomial for RAGs, or ordinal models for DGs). All models assume an additive relationship between punishment and omniscience on cooperation. Covariates in the main analysis include age, sex, education, number of children, material security, fisher occupation and game features. Sensitivity analysis 1 removes all covariates other than fisher occupation and game features. Point estimates are median values, with 95% credible intervals in brackets. Results are presented for both the multiple imputation results (MI; $n = 79$) and the complete-case analysis results (CCA; $n = 56$).

	Local RAG		**Self RAG**		**Local DG**		**Self DG**	
	MI	*CCA*	*MI*	*CCA*	*MI*	*CCA*	*MI*	*CCA*
Main analysis	0.24 (-1.71 to 2.20)	1.20 (-1.02 to 3.41)	-0.01 (-1.90 to 1.87)	-0.59 (-2.86 to 1.63)	-0.42 (-1.39 to 0.59)	0.89 (-0.30 to 2.20)	-0.39 (-1.57 to 0.82)	0.30 (-1.05 to 1.73)
Sensitivity analysis 1	0.20 (-1.63 to 2.09)	1.45 (-0.73 to 3.50)	-0.23 (-2.03 to 1.57)	-0.54 (-2.70 to 1.52)	-0.30 (-1.29 to 0.73)	0.79 (-0.43 to 2.09)	-0.43 (-1.57 to 0.73)	1.13 (-0.30 to 2.63)

5. Discussion

To recapitulate our main results, overall we found little convincing evidence that supernatural punishment and omniscience beliefs towards a local god (*zazavavindrano*) were associated, either positively or negatively, with cooperation towards distant *zazavavindrano* co-religionists among the Sihanaka. These patterns were largely consistent across all model specifications – i.e., RAG vs. DG, self vs. local conditions, different assumptions regarding confounding, imputed vs. complete-case analyses – although the local DG results were somewhat more ambiguous, with a larger range of interpretations possible, from weakly negative to weakly positive (but also plausibly null).

Previous cross-cultural work focusing on cooperation among predominantly "moralistic" religious traditions found a small but consistent positive relationship between supernatural punishment beliefs and broader religious cooperation (Lang et al., 2019, 2023; Purzycki et al., 2016, 2024). These results among the Sihanaka suggest that local *zazavavindrano* religious traditions may not play the same role of promoting cooperation with distant local co-religionists. One interpretation of this is that local gods may therefore be less effective at promoting cooperation than moralistic gods (Norenzayan et al., 2016), although such a conclusion must be tempered for a few reasons.

First, *zazavavindrano* are predominantly concerned with – and believed to punish for – a specific set of behaviours tied to their aquatic domain: pollution of water, disrespect toward their dwelling places, breach of relational pacts between humans and aquatic beings, and violations of taboos (*fady*) associated with the wetlands. In Sihanaka cosmology, *zazavavindrano* are metahuman beings – a population inhabiting submerged villages beneath Lake Alaotra – who guard the purity and order of their own realm (Norge, 2025). Their punitive capacity is exercised in response to transgressions against them and their domain, not against the norms governing cooperation among humans across distance (as reflected in the difference in cultural models, compared to Ndrianampanjaka; Figure 3). Given this, it is perhaps not surprising that *zazavavindrano* beliefs do not promote the type of cooperation assessed here. This does not necessarily mean that local gods more generally *cannot* expand the circle of cooperation to the same extent as moralistic gods, only that many local gods did not appear to have evolved to serve this function, while many moralistic gods – especially from "world" religions such as Christianity, Islam and Hinduism – may have (Bendixen et al., 2024). A fairer test would be to compare cooperation towards distant co-religionists between moralistic and local religious traditions whose deities are equally concerned with cooperative behaviour towards strangers. However, as different deities often have different concerns and serve different functional niches (Bendixen et al., 2024), such comparisons may be difficult to find, especially within the same population.

Second, these results do not compare "moralistic" and "local" religious traditions within the same population; we simply do not know whether results would have differed looking at

Ndrianampanjaka (the designated moralistic god here) beliefs on cooperation with distant Ndrianampanjaka co-religionists among the Sihanaka, compared to *zazavavindrano*. This comparison would be particularly instructive, since Ndrianampanjaka is strongly associated with moral conduct, social order, and the punishment of transgression within his territorial realm – a domain encompassing the Anororo area and its surrounds, a population of perhaps 15,000 people, not all of whom are personally acquainted. Ndrianampanjaka is thus moralistic in his concerns but territorially bounded in his reach, unlike the universal moral scope typically attributed to the "big gods" of world religions (Norenzayan et al., 2016). This territorial boundedness is not, however, inherent to non-transcendentalist traditions as such. Elsewhere in Madagascar, divinised rulers such as Andriamisara and other royal divinities of the Sakalava in western Madagascar are venerated by substantially larger populations, and their influence extends well beyond any single polity – reaching across Sakalava realms and even to possession cults on the East African coast (Ballarin, 2007; Larsen, 2008; Giles, 2018). These divinities operate within the same immanentist cosmological framework as Ndrianampanjaka, yet their effective reach approaches that of the prosocial traditions foregrounded by some formulations of the supernatural punishment hypothesis (Norenzayan et al., 2016). Such cases suggest that the cooperative scope of a deity may depend not on whether it is "moralistic" in its concerns, but instead on the scale at which it is venerated and the breadth of social relations it mediates. A question that deserves further attention, then, is what enables some deities to extend their reach across large and heterogeneous populations. After all, even the deities now at the centre of "world religions," Yahweh for example, were once local gods with circumscribed followings (Smith, 2002). An important part of the answer may lie in a specific ontological transformation: the move from immanentism – where divinities are situated beings anchored to specific places in the landscape (marshes, tombs, shrines) – to transcendentalism, where divine authority is removed from the cosmos altogether and located in a radical otherworld (Moin & Strathern, 2022; Sahlins, 2022; Strathern, 2019, 2024). It is this detachment from place and polity that may facilitate the kind of universal moral jurisdiction foregrounded by certain variants of the supernatural punishment hypothesis (Baumard & Boyer, 2013; Norenzayan et al., 2016) – and it is precisely this detachment that is absent in Sihanaka cosmology, where all divinities remain embedded in the landscape they inhabit and the populations they govern.

Third, the Sihanaka case illustrates a broader pattern that may be characteristic of immanentist cosmologies: the domain-specificity of divine authority. In such systems, divinities are situated agents who exercise their power within particular realms, over particular populations, and in response to particular kinds of transgression. Among the Sihanaka, *zazavavindrano* guard the aquatic realm, Ndrianampanjaka upholds social and ritual order within his territorial domain, and the *razana* (ancestors) oversee kinship morality within the descent group. This means that the moralistic/local distinction, while analytically productive for cross-cultural comparison, may in practice conflate two dimensions that are independent in immanentist cosmologies like that of the Sihanaka: the *content* of a deity's concerns (moral vs. non-moral) and the *scope* of its authority (universal vs. bounded). Ndrianampanjaka is broadly moralistic but bounded in

territorial scope; *zazavavindrano* are moralistic within an ecologically specific domain and likewise bounded. Attending to this distinction may help refine future comparative work on the relationship between religious beliefs and cooperation, particularly in traditions where divine authority is distributed across multiple situated agents rather than concentrated in a single omniscient deity.

Fourth, and most fundamentally, these results invite a reconsideration of what "prosocial" means in the context of immanentist cosmologies. The experimental games used here measure cooperation between humans – specifically, the willingness to share resources with anonymous, geographically distant co-religionists. Yet in Sihanaka cosmology – as in immanentist traditions more broadly, including animistic ones (Bird-David, 1999; Howell, 1989; Descola, 2014; Århem & Sprenger, 2016) – the relevant social field extends well beyond the human community. The wetlands surrounding Lake Alaotra are not "nature" in the Western sense of a domain external to society; they are inhabited places, populated by *zazavavindrano* who maintain their own villages, fields, and herds beneath the water. Relations between humans and the marshland are therefore social relations, mediated by obligations, taboos, and reciprocal expectations – a pattern characteristic of human–water being relationships cross-culturally (Strang, 2023). To pollute the water is not merely an ecological offence but a social transgression against co-inhabitants of a shared landscape; to respect the *fady* of the wetlands is to uphold a relational order that sustains both human and metahuman communities.

Zazavavindrano beliefs, in this light, do promote prosocial conduct – but "social" here encompasses the broader community of human and metahuman beings who share the landscape, rather than the narrower circle of human-to-human cooperation that our experimental measures were designed to capture. *Zazavavindrano* are also said to punish betrayal of trust in relationships between humans and *zazavavindrano* – such as when a human spouse of a *zazavavindrano* reveals the nature of the union or breaks a pact – suggesting that these beliefs enforce norms of fidelity and reciprocity within intimate human–metahuman relationships, rather than anonymous generosity toward distant strangers.

This reframing opens a particularly promising avenue for future research. If *zazavavindrano* beliefs shape moral conduct within the human–environment relationship, their cooperative effects may be detectable in domains that standard experimental games do not capture: compliance with ecological taboos governing the wetlands, sustainable fishing practices, restraint in the exploitation of marsh resources, or the management of water quality in shared waterways. Future research could test this by designing behavioural measures that capture cooperation in domains where *zazavavindrano* are understood to be active agents – for instance, resource-sharing dilemmas involving fisheries or wetland commons – and comparing these effects with those of beliefs in territorially moralistic deities such as Ndrianampanjaka. Such an approach would speak not only to the evolution of religion literature, but also to growing interest in the role of traditional cosmologies and sacred ecologies in environmental governance and

conservation. In a region where Lake Alaotra – Madagascar's largest freshwater body – faces severe ecological degradation from overfishing, invasive species, and the conversion of marshland to rice paddies (Lammers et al., 2015), understanding whether and how beliefs in aquatic divinities shape environmental behaviour is not merely of theoretical interest.

Crucially, rather than dismissing local gods as less capable of promoting cooperation due to a lack of cognitively-appealing features related to moral intuitions and motivations (cf. Baumard & Boyer, 2013; Norenzayan et al., 2016), this perspective considers that different gods may have evolved to solve different adaptive challenges – and that the adaptive problems solved by ecologically situated divinities such as *zazavavindrano* may be no less consequential than those addressed by "big" or "moralistic" gods, even if they are less visible to measures designed with universalistic religious traditions in mind (Bendixen et al., 2024; Purzycki, 2025c; Purzycki, Bendixen, et al., 2022; Purzycki & Sosis, 2022).

There are few key strengths of this research, including: i) it is among the earliest formal, quantitative studies to focus specifically on the role of *local* gods in promoting cooperative behavior (see also Apicella, 2018; Atkinson, 2018; Kundtová Klocová et al., 2022; Vardy & Atkinson, 2022, plus other papers in this volume); ii) we took an explicitly causal approach to make it clear to readers the assumptions we are making in order to infer putative causal effects, and how different assumptions may impact conclusions; iii) use of multiple imputation methods to account for missing data and potential selection bias; and iv) an ethnographically-informed approach to ensure that our data collection and interpretations were grounded in the local religious context.

That said, this study of course possesses a number of limitations. A first key limitation is that of sample size. Our sample size of approximately 80 participants may have been insufficient to detect small – but still potentially meaningful – effect sizes. For instance, our RAG results had 95% credible intervals in the range of plus or minus two allocations, while the DG results were plus/minus approximately one allocation. *A priori* power analyses suggested that sample sizes of 40 per condition would be able to detect differences in RAG cooperation from 50% (i.e., equal allocations) to 54% (approximately a one-token increase in cooperation to distant co-religionists) with 90% power. However, power can be impacted by several factors, such as variation in cooperative behaviour and supernatural beliefs, and their relations with confounding factors. Therefore, while the majority of our point estimates were close to zero, given this uncertainty we cannot rule out small effect sizes in either direction, which may nonetheless be meaningful when scaled up at the population level.

Second, even though we adopted a causal inference approach to *try* and estimate causal effects, we cannot claim that we have *estimated* the true causal effects. There are some potential biases which may limit such an interpretation (Bulbulia, 2024; Hernán & Robins, 2020). We enumerate them in the following. (1) *Unmeasured confounding*: Although our results were robust to

different assumptions regarding observed confounders, there is always the risk of unmeasured confounders biasing these associations. Potential examples of these include lines of descent based on being a former slave or social stratification, both of which we were unable to collect data on due to the sensitivity of these topics in Madagascar (as discussed above). However, the fact that our results were largely null arguably makes such a bias less plausible, although we cannot rule out "masking" effects of a true non-null relationship due to unmeasured confounding. (2) *Reverse causality*: Given the cross-sectional nature of the data we have to assume that religious beliefs cause cooperation, rather than *vice versa*. While these experimental measures of cooperation cannot cause religious beliefs, previous cooperative behaviour may cause both subsequent religiosity (e.g., if successful cooperation with co-religionists further cements beliefs) and behaviour in these games. While a qualitative judgement, we believe that this reverse causal pathway is likely to be weaker than our proposed estimand. Nonetheless, a longitudinal design with repeated cooperation and religious belief measures to rule out this bias would provide stronger evidence for a causal interpretation (Bulbulia, 2024). (3) *Selection bias*: There may have been selection bias due to missing data in our sample, and though we applied methods to try and overcome this via multiple imputation, selection bias can also occur due to non-representative sampling (Lu et al., 2022). We believe this risk of bias is minimal as our sample is broadly representative of our target population. (4) *Measurement error*: We have to assume that all relevant variables – *zazavavindrano* beliefs, cooperation with distant *zazavavindrano* co-religionists, and all confounders – are measured without error, else our causal estimates may be biased (Greenland, 1980; van Smeden et al., 2020). Most confounders are easy to measure (e.g., age, sex, number of children) so are at less risk of bias (and, even if they were measured with error, as most appear to not actually confound our focal estimand, any bias is likely to be minimal). RAG and DG behaviours are not measured with error (although whether they reflect real-world cooperation and are unaffected by demand characteristics – i.e., ecological validity - is another matter). Beliefs – especially religious beliefs – are much harder to measure accurately.

Most fundamentally, the standardised question format presupposes a framework of "believing in" divinities that sits uneasily with Sihanaka religious epistemology, where the operative concept is less *mino* ("to believe") than *manompo* ("to serve"). Many Sihanaka do not relate to *zazavavindrano* as objects of propositional belief but as beings with whom one maintains – or does not maintain – relationships of service and obligation. A question such as "Do *zazavavindrano* ever punish people for their behaviour?" thus arguably translates a relational practice into a cognitive attitude, which may not capture how participants actually orient toward these beings. A further consideration concerns the social dynamics of disclosure. In rural Madagascar, modernity, formal education, and scientific authority have long been associated with Christianity, which since its introduction in the nineteenth century has at times denigrated indigenous ritual practice as backward or demonic. This hierarchy is occasionally internalised by Sihanaka themselves, and may produce reticence when discussing traditional cosmological commitments with outsiders – particularly when the lead researcher is European and the research

assistant is from the capital, Antananarivo. That said, this study builds on a decade of collaborative work in Anororo (since 2015), and many participants were well acquainted with the research team, which likely mitigated such effects considerably. Nevertheless, some degree of social desirability bias cannot be excluded. Finally, knowledge of *zazavavindrano* is not uniformly distributed within the community. Spirit mediums and ritual elders possess specialised knowledge of aquatic beings that ordinary villagers may not claim, even if they regularly participate in rituals where *zazavavindrano* possess mediums. Several participants who were known to attend such ceremonies nonetheless responded that they did not know about *zazavavindrano*, suggesting that the question was heard not as asking about general familiarity but about authoritative knowledge – a domain reserved for specialists. This uneven distribution of ritual authority may have contributed to patterns of non-response in the religiosity variables, and by extension to the missing data addressed through our imputation procedures. While such complexities are inherent to cross-cultural research using standardised instruments, they warrant acknowledgement when interpreting null results. That said, if the true causal relationship with cooperation is null, then any measurement error in our exposures is likely to be benign. Nonetheless, despite these risks of bias we hope that we have been transparent enough in our assumptions that readers can decide for themselves whether a causal interpretation is warranted (or, conversely, how much these potential biases may impact and question any causal interpretation, and the direction of any potential bias).

In sum, there appears to be little relationship between *zazavavindrano* punishment and omniscience beliefs and the expansion of cooperation with distant co-religionists of this local religious tradition among the Sihanaka. Rather than interpreting this as evidence that local gods are simply less effective at promoting cooperation, however, the ethnographic context points toward a more expansive understanding of what religious beliefs may do. In Sihanaka immanentist cosmology, divine authority is characteristically situated: *zazavavindrano* regulate the aquatic domain and the relational obligations that bind humans to their wetland environment, while Ndrianampanjaka enforces social and ritual order within his territorial realm. Neither deity's concerns map onto the anonymous, distance-spanning cooperation that our experimental games were designed to measure – but this does not mean they lack prosocial effects, even in ways that expand beyond parochial borders. Rather, it means that the social field they govern is differently constituted: in one case territorially bounded, in the other ecologically situated, and in both cases encompassing relations that extend beyond the human community. A promising line of future research would therefore be to move beyond the question of whether local gods expand the cooperative circle among humans, and instead investigate whether beliefs in situated deities shape behaviour in the broader relational fields those deities actually govern – including ecological stewardship, resource management, and the observance of environmental taboos (cf. Hames, 2007; Smith & Wishnie, 2000). Such an approach would not only do greater justice to the cosmologies in which these beliefs are embedded, but may also reveal that the prosocial reach of "local" religious traditions has been underestimated – not because they fail to promote

cooperation, but because the cooperation they promote has not yet been measured. We are currently pursuing these possibilities.

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Authors' roles
AN collected data, DM-S and BGP managed the project and contributed to analysis, and BGP designed the protocols. All authors contributed to writing this manuscript.

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Data availability statement
All data and analysis code will be made available on the project GitHub website: https://github.com/bgpurzycki/GGSL-Project/tree/main/Code/Objective%201/Sihanaka.

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Appendix

Table A1: Summary of free-list coding scheme.

Name	Description
Morality	Generalized behaviors that have a benefit or cost to other people (e.g., hurting, generosity, sharing, etc.)
Virtue	Individual qualities that may or may not have social ramifications (e.g., hard-working, patient, bad conscience, etc.)
People	In reference to the quality and/or state of people (e.g., people, people stay in good health, happy, etc.)
Etiquette	Conventional social behaviors that have no immediate cost or benefit to others (e.g., shaking hands, wearing the proper clothes, etc.)
Substance Use/Abuse	Items that involve the use of illicit substances (e.g., alcohol, smoking, drugs, etc.)
Religion	Any non-ritual or non-behavioral item concerned with the supernatural (e.g., faith, devotion, belief, etc.)
Ritual	Any behavior or object used in ritual devoted to the supernatural (e.g., prayer, meditations, offerings, sacrifices, etc.)
Ecology	Any behavior or object affecting non-human relationships (e.g., pollution, gardening, wild plants/animals, etc.)
Food	Food items (e.g., yam, milk, etc.)
Miscellaneous	Miscellaneous items that cross-cut categories or do not fit into any category
D/K	Responding “I don’t know” or “Not sure,” or similar
Specific	Items that are specific to a culture (e.g., bel’ leaf, *artysh*, etc.; these were usually coded into one of the above categories, after consultation with field researchers)